

Business Analysis

With a minimum Support level of 0.01, a minimum Confidence level of 0.50, and a minimum length of 2, 98 rules were produced using the Apriori algorithm across a total of 33,541 transactions.

This analysis offers multiple opportunities to upsell and cross-sell products. The top ten rules are listed in the accompanying PDF. Key rule insights and practical retail applications focused on five sets of combinations:

1. Customers who bought Poppy's Playhouse Living Room also bought Poppy's Playhouse Kitchen 89.3% of the time. Customers who bought Poppy's Playhouse Kitchen also bought Poppy's Playhouse Living Room 66.1% of the time. Customers buying one are 56 times more likely to buy the other.
2. Customers who bought Poppy's Playhouse Kitchen also bought Poppy's Playhouse Bedroom 75.9% of the time. Customers who bought Poppy's Playhouse Bedroom also bought Poppy's Playhouse Kitchen 85.4% of the time. Customers buying one are 54 times more likely to buy the other.
 - a. These first two combinations offer a great upselling opportunity. Customers buying the Kitchen were very likely to buy the Living Room or the Bedroom. The store can bundle all three of these items to upsell the customer. Cross-selling can also be achieved by placing these items next to one another in the store.
3. Customers who bought the Set-6 Red Spotty Paper Cups also bought the Set-6 Red Spotty Paper Plates 77.4% of the time. Customers who bought the Set-6 Red Spotty Paper Plates also bought the Set-6 Red Spotty Paper Cups 66.3% if the time. Customers buying one are 51 times more likely to buy the other
 - a. This combination offers a great cross-selling or up-selling opportunity. Cross-selling could be done by ensuring the cups are located close to the plates in the store. Or the store could run a special on a bundle of cups and plates to upsell.
4. Customers who bought the Blue Stripe Ceramic Drawer Knob also bought the Red Stripe Ceramic Drawer Knob 61.1% of the time. Customers who bought the Red Stripe Ceramic Drawer Knob also bought the Blue Stripe Ceramic Drawer Knob 66.5% of the time. Customers who buy one are 39 times more likely to buy the other.
 - a. This combination offers a great cross-selling opportunity. These are likely already placed in close proximity in the store. If they aren't, then they should be to cross-sell the customer.
5. Customers who bought the Key Fob, Back Door also bought the Key Fob, Shed 72.5% of the time. Customers who bought the Key Fob, Shed also bought the Key Fob, Back Door 63.7% of the time. Customers who buy one are 38 times more likely to buy the other.
 - a. This combination offers a great cross-selling opportunity. Again, these are likely already placed in close proximity to one another in the store. If they aren't, then they should be to cross-sell the customer.

All of the aforementioned combinations can be used to cross-sell to customers, increasing the number of items in their baskets. The cups/plates combination and the playhouse combinations offer a strong opportunity to up-sell these items together. Cups and plates each come six to a pack and could be advertised for cookouts or any family gathering where the host doesn't want to wash dishes. The playhouse combinations could easily be an upsell for Christmas time.